



ENGO 545 – Fall 2026 Hydrographic Surveying Laboratory exercise #1 (part 2) Safety of navigation



Deadline: Mon, Oct 5, 2026 (11:59 PM)

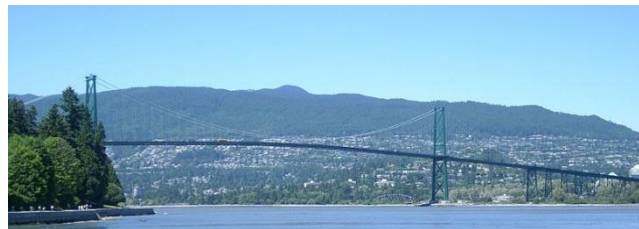
Learning objectives:

After completing this laboratory exercise students should be able to:

- Identify features on a nautical chart pertaining to the safety of navigation
- Apply different water levels and tidal datums in the context of depths and elevations shown on a nautical chart
- Conduct tide prediction for a Canadian tidal station
- Write a technical report

Background information:

You are captain of a large cruise ship. The draft of the ship is 9.8 m and the superstructure of the ship extends 61.5 m above the water line. You are asked to make a port visit in Vancouver, transiting through the First Narrows under the Lion's Gate Bridge. Assume the vessel requires at least 5 m below keel clearance and 3 m height clearance for safety purposes.



Hints:

a) Find Vancouver Harbour tides at the Canadian Hydrographic Service (CHS) website:

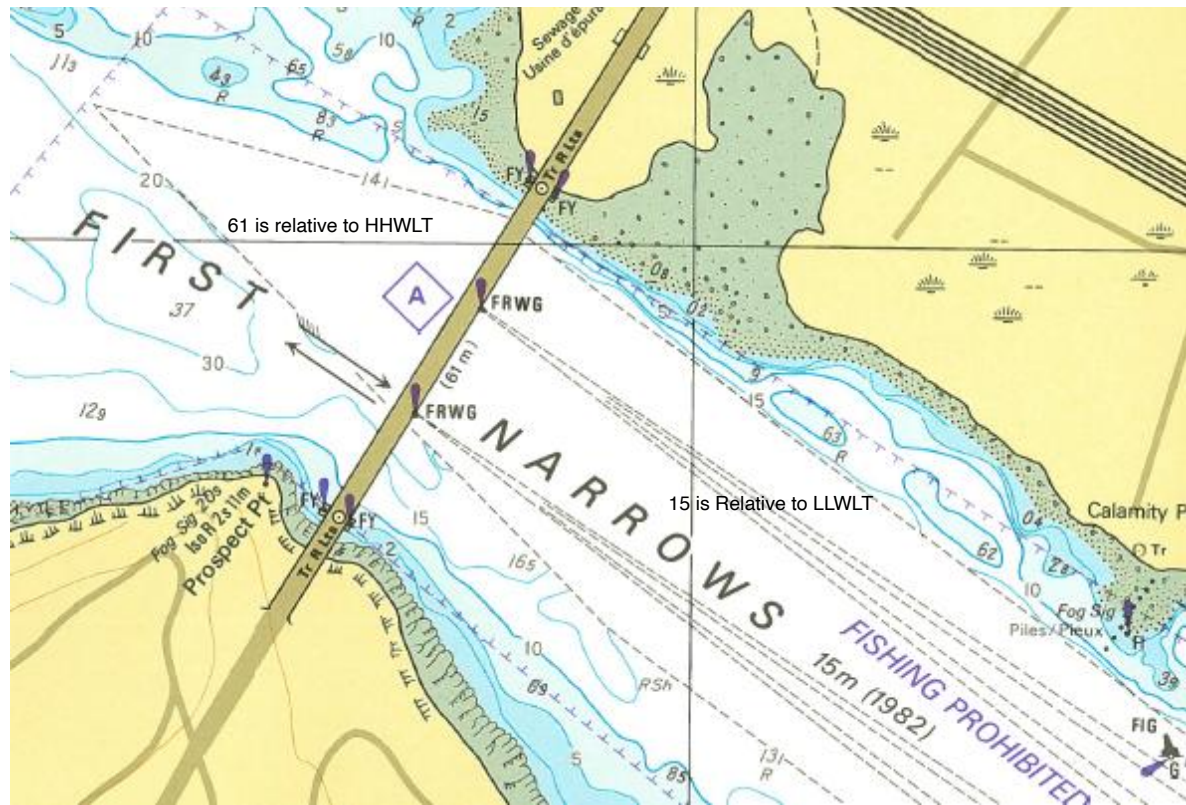
<http://www.waterlevels.gc.ca/eng/station?sid=7735>

b) Use the following water level values for the Vancouver reference station (taken from the 2017 Canadian Tide and Current Tables, Volume 5 – Juan de Fuca Strait and Strait of Georgia):

Station ID 7735-Vancouver (values in metres)						LLWLT is datum for depth			
Tide Type	HHWLT	HHWMT	MWL	LLWMT	LLWLT	Large Range	Mean Range	Extreme High	Extreme Low
MSD	5.0	4.5	3.1	1.2	0.1	4.9	3.3	5.6	-0.3

HHWLT is the datum for height

Excerpt from a Nautical Chart (First Narrows can be seen on nautical charts 3481 – Approaches to Vancouver Harbour, and 3493 – Vancouver Harbour Western Portion published by the Canadian Hydrographic Service in 2012)



Tasks:

$x > ?$ $x < ?$

- Given the draft of the vessel, what tide height (or higher) will clear the keel at the shallowest depth? Similarly, given the height of the superstructure, what tide height (or lower) will clear the superstructure under the bridge? *Hint: use inequalities to solve these problems.*
- Given the two-week period from Mon, Nov 2, 2026 to Sun, Nov 15, 2026, what time in the morning can you enter and what time in the evening can you depart the harbour on each of the 14 given days? Use a spreadsheet or source code to answer this question. Present your answers in a table and/or graphically.
- From the 14 given days, which day(s) (if any) would you recommend to make the port visit in order to maximize the time the passengers have to sightsee the city of Vancouver? State any assumptions.

Deliverables:

Check the Additional Course Information document on D2L for the required deliverables on a laboratory exercise.